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Artificial Vision for the Blind by Electrical Stimulation of the Visual Cortex

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Artificial vision for the blind may be feasible by interfacing a television camera with electronics stimulating the visual cortex. The status of a major collaborative effort involving the College of Physicians and Surgeons of Columbia University, the University of Utah, and the University of Western Ontario is reviewed. Results have been very encouraging, although much work remains to be done.

Key words: Artificial vision, Electrical stimulation, Sensory aids, Visual cortex, Visual prostheses

INTRODUCTION

Benjamin Franklin suggested the idea of using electricity to restore sight to the blind and hearing to the deaf when he described his experiments on lightning to the Royal Society of London (23). Before the end of the 18th century Franklin, Volta, LeRoy, and others had begun experiments on blind and deaf subjects using crude external electrodes.

Fragmentary reports of phosphenes (punctate sensations of light) produced by direct electrical stimulation of the visual cortex were first published in the late 1920s (10, 17). If the electrode was small (e.g., 1 mm^2), the sensation was that of a small spot of light at a specific location in visual space; if the electrode was moved to a new cortical location, the phosphene appeared to move to a new location in space.

An additional report by Penfield and his associates of phosphenes produced by electrostimulation resulted from their experiments during resection of lesions under local anesthesia, but in many cases this apparently involved facilitation of epileptic aura (21). Although a few researchers subsequently speculated that it might be possible to use phosphenes as the basis for a visual prosthesis (2, 18, 22), the concept did not receive wide attention until 1968 when Brindley and Lewin conducted a pioneering experiment in which 81 subdural electrodes were implanted in a blind former nurse, with associated circuitry between the scalp and skull (1). Their report confirmed that small reproducible phosphenes could be elicited in a totally blind subject, encouraging research by our group and others.

The objective of this research is the creation of visual images composed of appropriately chosen sets of discrete phosphenes, analogous to the images created by illuminating selected sets of lights on a scoreboard. The quality of such images is greatly enhanced if each phosphene is made to vary in brightness, and this can be accomplished easily during visual cortex excitation by altering various stimulus parameters. Blind subjects might eventually be equipped with a portable system consisting of a tiny TV camera, a microcomputer to analyze the visual scene and control stimulation, and an array of implanted electrodes. This goal is illustrated in Figure 1.

Independent mobility is much more important than reading to most blind people, particularly because tape recordings and other alternate sources of printed information are available. Especially since World War II, numerous efforts have been made to develop noninvasive aids for the blind based on the conversion of visual information for detection by touch or hearing (13). Although well known to the blind and to professionals involved in their rehabilitation, these devices are not widely used because they are expensive, transmit information very slowly, and are difficult to learn to use. One example is a sonar system to help detect obstructions while walking; subjects must learn to relate changes in sound output to the distance and location of objects in space. This is very difficult, and similar difficulties contribute to the fact that fewer than 20% of the blind can read braille, fewer than 10% can use a cane for traveling, and fewer than 1% can use a guide dog successfully.

Electrical stimulation of the visual cortex has the advantage of directly activating the visual system so that no new "code" must be learned. Also hearing is left free, an important advantage because blind people depend heavily on auditory cues. Of course, this system requires an intact visual cortex, but because the damage is peripheral in the vast majority of cases of blindness this requirement poses no serious problems. Recently, very encouraging work on computerized systems that actually "read" text aloud has been reported (R. Kurzweil, personal communication, 1979), but this approach is not applicable to mobility, which has been our principal interest.

HISTORY OF THE PROJECT

Our research on visual prostheses began at the University of Utah in collaboration with the University of Western Ontario and is now continuing at the Columbia-Presbyterian Medical Center in New York City. To explore artificial hearing for the deaf, analogous experiments by our group have involved stim-

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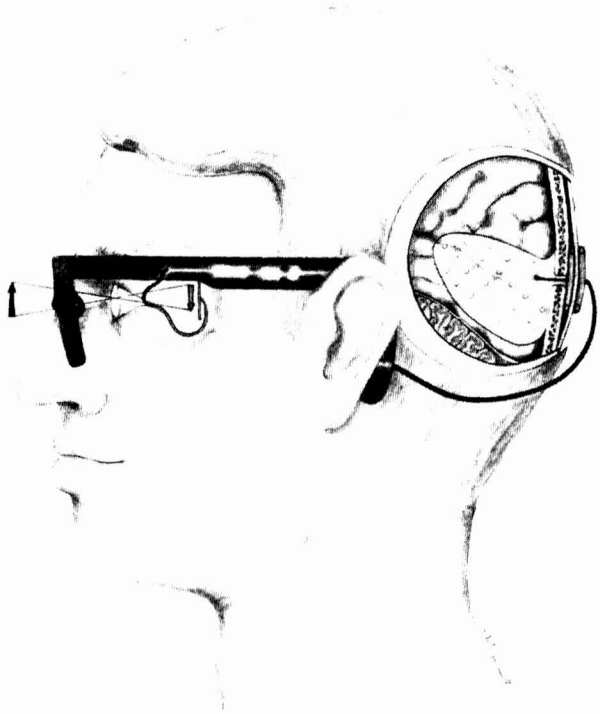

FIG. 1. Artist's conception of a future visual prosthesis for the blind. A subminiature television camera implanted in the orbit is connected to electrodes stimulating the visual cortex. The electronics are controlled by a microcomputer mounted in the frame of a pair of dummy glasses.

ulation of the auditory cortex (6). Although these initial temporal lobe experiments were encouraging, the emphasis has been shifted to intracochlear stimulation (8). Many sensorineural deaf patients have residual 8th nerve and the surgery required to implant electrodes in the cochlea entails less risk than the surgical exposure of Heschl's gyri, which are buried deep in the posterior portion of the sylvian fissure.

To minimize surgical risks while refining the instrumentation and techniques for visual cortex stimulation, we conducted an initial series of experiments on sighted patients who were undergoing operation for the removal of tumors or other lesions of the occipital lobe. Because surgery involving exposure of the visual cortex under local anesthesia is relatively uncommon, a program was organized with the collaboration of approximately 60 neurosurgical services throughout the United States and Canada. This enabled our group to complete stimulation experiments on 37 patients during 4 years beginning in 1969 (3).

By 1973 our group had sufficient experience to conduct experiments using temporary implants in otherwise healthy blind volunteers (5). These experiments demonstrated recognition of simple patterns such as triangles, squares, and a few letters. However, it was clear that further progress required chronic implantation in blind volunteers.

Before taking the next step, a decision had to be made about the electronic equipment needed for a chronic implant. Was it better to place an elaborate electronics package under the scalp, as Brindley had done, and use radiofrequency coupling to transmit information? Or should all of the electronics be external to the body?

The second alternative, which meant bringing wires through the skin, was chosen because it offered major advantages. This method eliminates cross talk between channels, which commonly occurs with inductively coupled systems, and allows

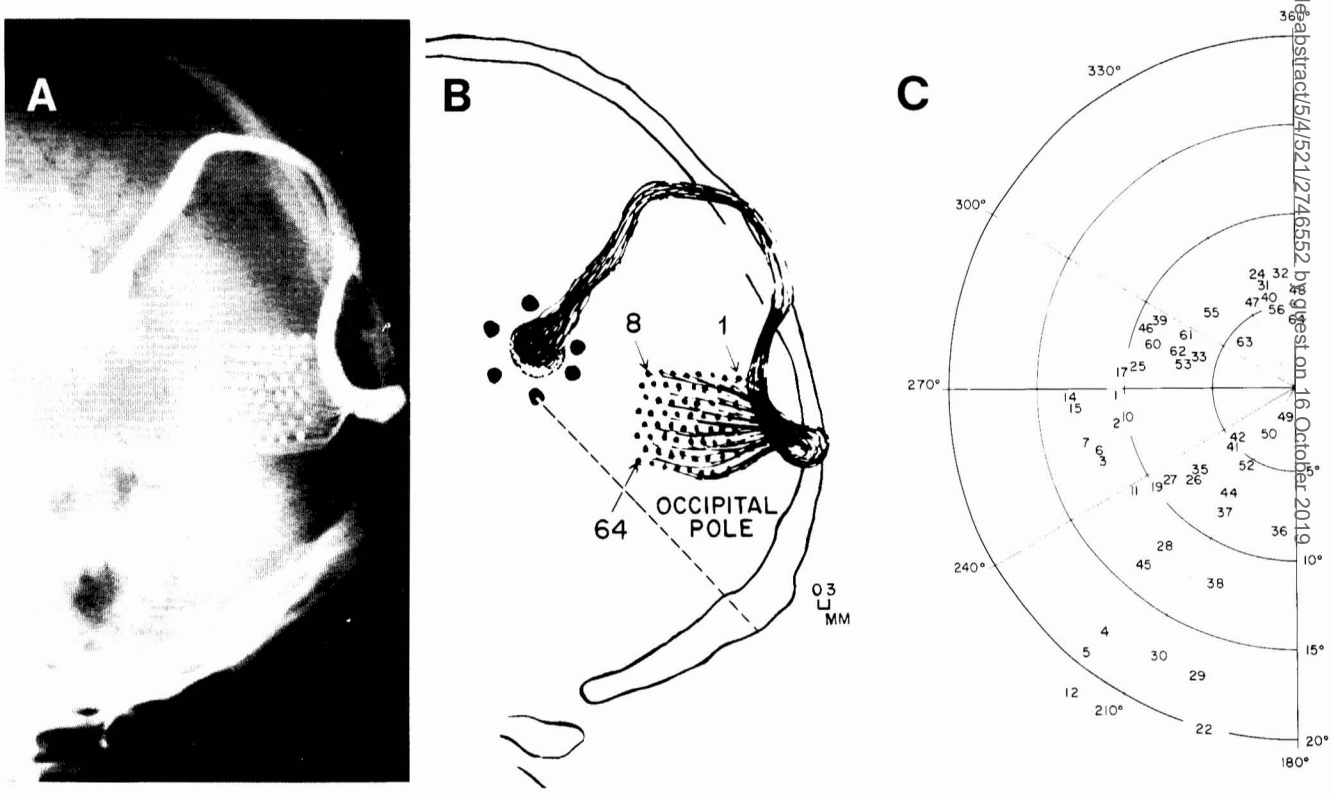

FIG. 2. Electrodes implanted against the right medial occipital cortex of a volunteer subject. The bundle of connecting wires passes through the skull, tunnels beneath the periosteum, and attaches to a percutaneous connector, which emerges through the skin behind the right ear. *A*, lateral x-ray film. *B*, artist's rendition showing the electrode numbering convention; *dashed line*, approximate location of the tentorium. *C*, map showing the position of the phosphene elicited in the visual field by each electrode.

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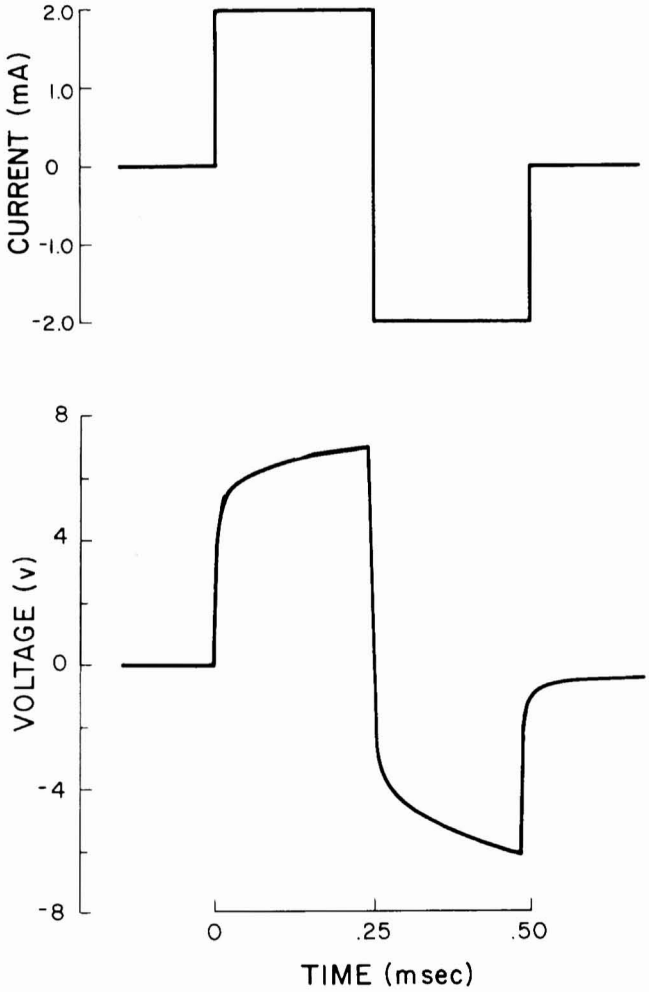

FIG. 3. Stimulus waveform used for most experiments. The amplitude is typically 2 ma (0 to peak) and the pulse duration is typically 0.25 msec/phase. The typical electrode impedance is 2.5K ohms (e.g., 5 volts to produce 2.0 ma).

precise determination of the amount of current applied to each electrode. An external electronics package also allows refinement of the stimulation system based on the experimental results and is more reliable and easier to repair. Also, our group had just developed a similar connector to bring wires through the skin for use in parallel experiments involving chronic scala tympani implants in deaf volunteers (19). These cochlear stimulation experiments suggested that the risk of infection would be minimal if a somewhat larger version of this new percutaneous connector was used to bring wires through the scalp in blind volunteers.

INSTRUMENTATION

The button-shaped connector, or pedestal, that was subsequently fabricated for use in visual cortex stimulation experiments is made of a noncorrosive, nontoxic type of carbon. The main flange lies under the scalp and is attached to the skull with bone screws. The tubular portion protruding through the skin behind the right ear has a diameter of approximately 1 cm, is hidden by the patient's hair, and causes no discomfort. In fact, patients often remark that they are unaware of its presence (16).

Figure 2 shows the electrode placement and resulting phosphene map. The array is subdural and is placed against the visual cortex of the right hemisphere. Individual electrodes are 1-mm^2 platinum discs, with centers 3 mm apart, embedded in a thin Teflon strip (15). Teflon-insulated leads pass through the dura and cranium and terminate in the special connector. An external ground is usually used, but other experiments in which one or more other electrodes serve as ground have also been conducted.

Extraordinary precautions to ensure sterility are undertaken during the operative procedure. The cable connecting the electrode array to the pedestal is tunneled a substantial distance beneath the galea and periosteum. A cable from the external electronics can then be plugged into the connector during experiments. The stimulator (seen in Fig. 10) is computer-

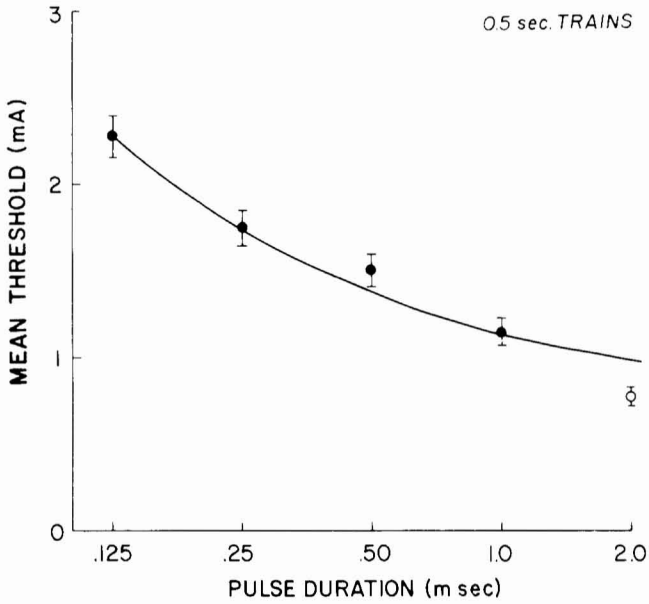

FIG. 4. Threshold vs. pulse duration for a 0.5-second train of pulses. (From J. P. Girvin, J. Evans, W. H. Dobelle, M. G. Mladejovsky, D. Henderson, I. Abramov, J. Gordon, and J. Turkel: Electrical stimulation of human visual cortex: The effect of stimulus parameters on phosphene threshold. Sens. Processes. In press.)

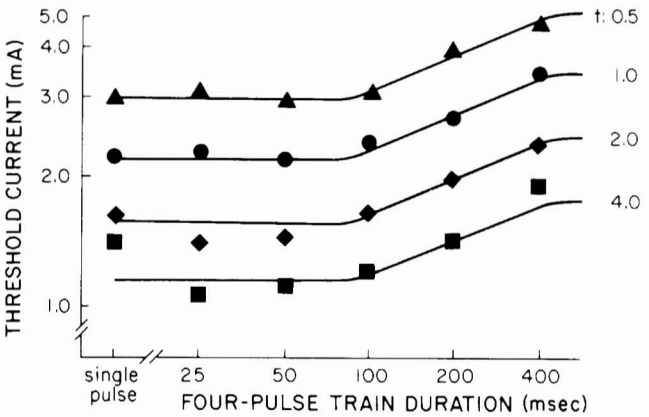

FIG. 5. The threshold for four-pulse trains increases substantially when they are spread over a period longer than 100 msec. Repeating the experiment with different pulse durations shifts the mean threshold, but the same 100-msec inflection point is seen for trials with the four-pulse durations plotted in this figure. (From D. Henderson, J. R. Evans, and W. H. Dobelle: The relationship between stimulus parameters and phosphene threshold/brightness, during stimulation of human visual cortex. Trans. Am. Soc. Artif. Intern. Organs. In press.)

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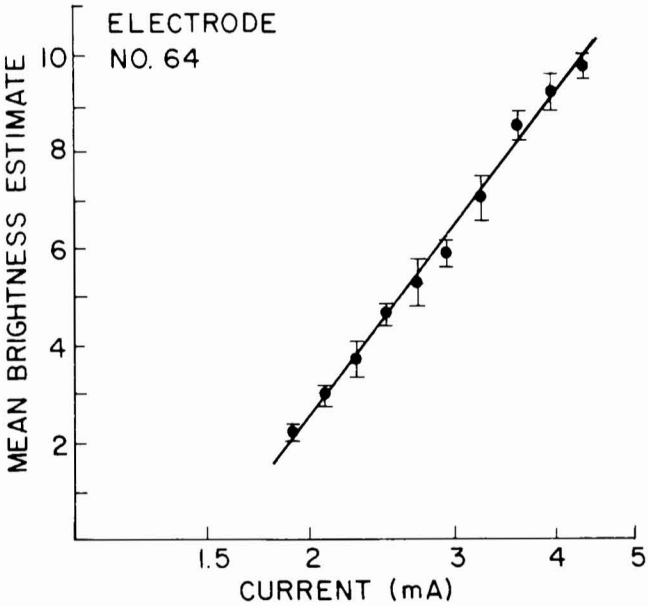

FIG. 6. Phosphene brightness as a function of stimulus current for a representative electrode. Brightness was measured by direct magnitude estimation. Each *point* is the mean of 10 estimates. (From J. Evans, J. Gordon, I. Abramov, M. G. Mladejovsky, and W. H. Dobelle: Brightness of phosphenes elicited by electrical stimulation of human visual cortex. Sens. Processes. In press.)

controlled, produces constant current pulses, is capacitively coupled to the electrodes, and is optically isolated from AC line power (20). This method of percutaneous connection affords full experimental flexibility, and problems with failure of the implanted electronic components have been eliminated.

VOLUNTEERS

A rigorous program for informed consent has been followed throughout the experiments with blind and deaf volunteers. Psychiatric screening, psychometric testing, and interviews are used to ensure a thorough understanding of the project, its risks, and the potential complications by the volunteers and their spouses (14). Major agencies for the blind collaborated in all initial searches for prospective blind volunteers, concentrating on emotionally stable, independent blind persons rather than dependent "clients." Volunteers for visual cortex stimulation have been restricted to totally blind adults with acquired handicaps to ensure the ability to describe visual sensations from memory. The ability to cooperate in prolonged experiments is obviously very important.

There have been three temporary implantations in blind volunteers, and eight additional volunteers (four blind and four deaf) have been given implants with chronic percutaneous connections. The two most recent visual cortex implantations were performed in December 1978, and the longest chronic cochlear and visual cortex implantations were performed over 4 years ago. Over 20 patient-years of experience have been accumulated without infection or other direct complications (16). The longest term visual cortex implant is in a man blinded by a gunshot wound that severed both optic nerves. The implantation was performed in 1975, when he was 33 years old (after 10 years of blindness), and experiments with him are still in progress. When he volunteered for the experiment he and his spouse were fully aware that there was no foreseeable chance that he would benefit directly from the implant; he derives no income from being a subject.

EFFECTS OF VARIOUS STIMULUS PARAMETERS

Stimulation typically involves trains of constant current, rectangular, biphasic pulses, as seen in Figure 3. The effects of the following stimulus parameters have been examined: (*a*) polarity, using either monophasic or biphasic symmetrical pulses; (*b*) current amplitude, measured 0 to peak, varied from 0 to 8 ma; (*c*) pulse duration, varied from 0.125 to 2.0 msec; (*d*) frequency, ranging from 12.4 to 1600 pulses/sec; and (*e*) train duration, from a single pulse up to trains of pulses 2.0 sec in duration.

Changing stimulus parameters affects phosphene brightness, but does not change the phosphene's shape, color, or location in visual space. Consequently, the effect of varying these parameters on the current required to elicit a just barely visible ("threshold") phosphene was systematically studied. Each electrode was tested separately using a "standard" set of parameters, and only one parameter was varied at a time (11). For example, the effect of varying pulse duration on threshold current was studied with 0.5-second trains consisting of 25 biphasic (+/−) pulses, as well as with single biphasic pulses. The tradeoff between pulse duration and pulse amplitude is

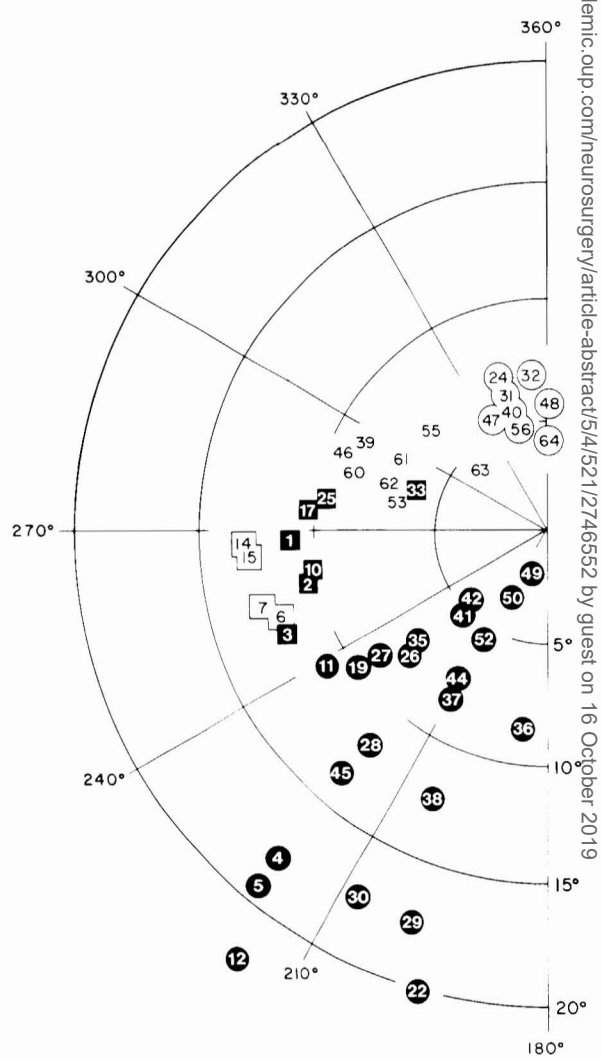

FIG. 7. The map of phosphenes in the visual field coded according to the portion of the cortex from which they were elicited. (From W. H. Dobelle, J. Turkel, D. Henderson, and J. Evans: Mapping the representation of the visual field by electrical stimulation of human visual cortex. Am. J. Ophthalmol. In press.)

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shown in Figure 4. If each phase is 2.0 msec long, about 1 ma is required for threshold. If the pulse duration is reduced to 0.25 msec, almost 2.0 ma are needed.

More recently, a special relationship has been found between pulse duration, the number of pulses in a train (which is a function of frequency), and the resulting sensation of light. It seems that, for any given pulse amplitude, the cortex integrates the charge delivered during the first 100 msec (12). Consequently, a single 1.0-msec pulse has the same effect as two 0.5-msec pulses or four 0.25-msec pulses, providing that all pulses are delivered during the 100-msec critical period. However, this tradeoff between phosphenes and the charge delivered apparently does not extend beyond 100 msec, as shown in Figure 5. Also, it holds true only for a fixed pulse amplitude. Doubling the amplitude while halving the pulse duration also results in charge constancy, but the phosphene thresholds and brightness are very different, although the charge is equal.

Electrode impedances and stimulation thresholds have remained constant over the 4 years of experimentation with our long term subject. No adverse effects of stimulation have been observed.

Brightness

Variation in brightness has been examined as a function of various stimulus parameters. Brightness increases with pulse amplitude (current) (Fig. 6). In general, brightness is related logarithmically to current amplitude, based on a magnitude estimation procedure in which the subject gives numerical ratings for perceived brightness (9).

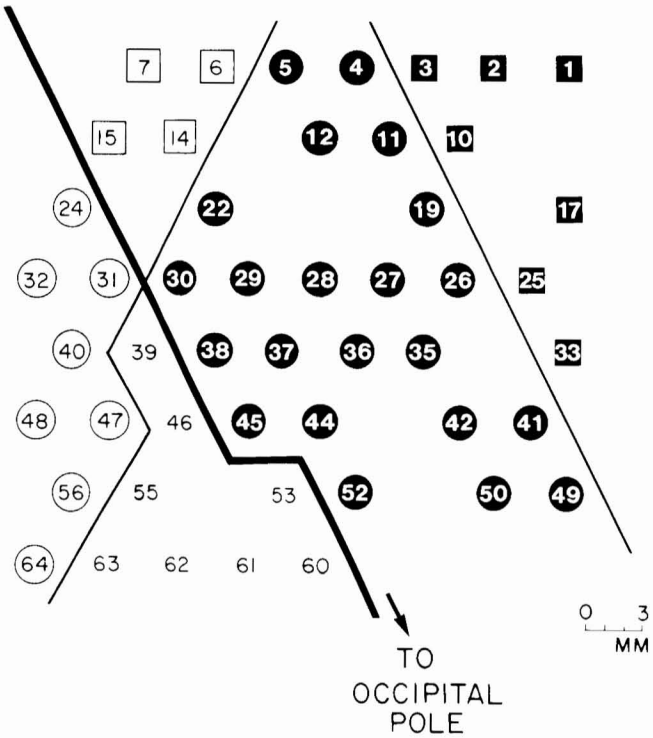

FIG. 8. Map of the electrode array corresponding to the phosphene map seen in Figure 7. The two boundaries running from *upper left* to *lower right* are thought to be sulci. The other boundary, running from the *middle* of the *upper edge* to the *lower left corner*, is probably a boundary between Areas 17 and 18. All 57 electrodes that had continuous wires produced phosphenes, but those that produced multiple phosphenes at threshold were eliminated from this analysis. (From W. H. Dobelle, J. Turkel, D. Henderson, and J. Evans: Mapping the representation of the visual field by electrical stimulation of human visual cortex. Am. J. Ophthalmol. In press.)

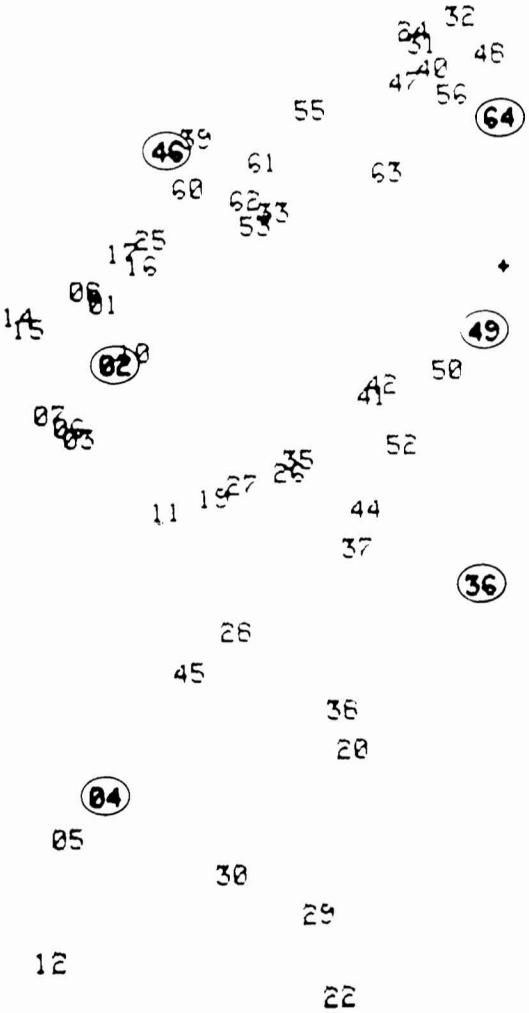

FIG. 9. The six circled phosphenes comprise a 3 × 2 braille cell, which has been used to transmit written information to the patient. (From W. H. Dobelle, M. G. Mladejovsky, J. R. Evans, T. S. Roberts, and J. P. Girvin: "Braille" reading by a blind volunteer by visual cortex stimulation. Nature, 259: 111–112, 1976.)

There is some question with all magnitude estimation procedures of whether the subject is scaling the desired sensation appropriately. Consequently, this direct scaling of brightness was validated by cross modality matching, in which hearing was used for comparison. The subject, who had normal hearing, was stimulated to produce a cortical phosphene and then asked to adjust the intensity of white noise (heard through earphones) until the loudness "matched" the brightness of the phosphene. The theoretical curve for such cross modality matching corresponded very closely to the actual data points. This indicated that the subject was scaling brightness properly and that it was possible to vary stimulus amplitude to achieve an adequate range of discriminable brightness.

Mapping

Each point in the visual field projects to a corresponding point on the visual cortex. This retinotopic relationship between electrode position and the location of the elicited phosphene in visual space is of particular importance in determining the position of sulci and other boundaries, such as that between Areas 17 and 18. It is difficult to visualize the mesial surface of the right occipital lobe with the patient lateral under local

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anesthesia, and determination of cortical boundaries would otherwise require multiple biopsies.

One of the problems when mapping phosphenes is that their location in visual space changes with eye movements. To eliminate this problem the subject is asked to "look" at his thumb, which is placed at the fixation point of a special electronic tangent screen. He then indicates the position of each phosphene relative to the fixation point by touching the appropriate position on the screen. In addition, a computer-controlled series of comparisons can be made between pairs of phosphenes (17); relationships such as "above and to the left" or "below and to the right" are then described.

Combining the results of both methods, a map can be derived, as shown in Figures 2 and 7. Each number shows where the phosphene elicited by a given electrode appears relative to the fixation point. Note that all of the phosphenes are in the left visual field, which is to be expected because the electrodes are on the right visual cortex.

Electrodes close together on the cortex may produce widely separated phosphenes because adjacent cortex corresponding to intermediate points in the visual field is buried in a sulcus and therefore is not accessible to surface stimulation (Fig. 8). In addition, electrodes far apart on the cortex may produce adjacent phosphenes by stimulating different cortical areas (such as Areas 17 and 18) with overlapping visual field maps. However, the details of such analyses are complicated and are discussed elsewhere (7).

TRANSMISSION OF INFORMATION FOR READING OR MOBILITY

The key question is whether useful images can be transmitted by stimulation of appropriately selected sets of electrodes. Consider the circled phosphenes in Figure 9. The subject, without training, had no trouble recognizing these as a braille cell (braille is a code of dots in a 2 × 3 matrix). By using these phosphenes the subject could, without practice, read braille 5 times faster than he could by using his fingers (4). This clearly demonstrated that visual images could be perceived by a subject without special training and at a useful rate of speed. However, there is no intention of using this approach to transmit braille for routine use.

More important, the subject was able to use a TV camera to detect simple horizontal and vertical lines (Fig. 10). However, when these experiments were done several years ago, we did not know enough about the effects of various stimulus parameters, brightness modulation, and phosphene mapping to pro-

FIG. 10. Volunteer using a television camera to detect horizontal and vertical lines. Such simple edge detection is fundamental to the development of mobility aids, although the large computer-controlled stimulation system in the background is obviously not portable.

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ceed. New information transfer experiments are now in progress based on work reviewed in this article.

FUTURE DIRECTIONS

In December 1978, two additional blind subjects received implants at the Columbia-Presbyterian Medical Center in New York. Experiments in these volunteers have been concentrated on confirmation of the major findings from the long term implantation experiment that are reviewed in this article. One of the recent volunteers lost his sight in an accident at the age of 6 (presumably after the visual cortex had fully developed) and was blind for 57 years at the time of implantation. Experiments with this subject have concentrated on possible changes in the visual cortex (neuronal plasticity) after acquired blindness.

The problems involved in selecting and rehabilitating appropriate candidates for visual prostheses may prove even more difficult to solve than the technological problems involved in developing these devices. However, such problems are far in the future. In the interim, new implantations in additional patients are planned. These will involve much larger arrays of electrodes (at least 256 per hemisphere), and eventually implants may be applied to both occipital cortices to double the number of electrodes. Future advances will depend on these purely experimental implantations in well-trained and highly motivated blind volunteers.

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